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11/30/2025

Overview / Goal

The goal of this project is to analyze data from the **NC State Women's Volleyball** team and build models that can help predict whether the team wins or loses a match based on performance statistics. The data used in this project includes the **2024 season** and the **2025 season through October 11**. Matches from October 17 through November 29, 2025 form a chronological holdout test set used to evaluate out-of-sample model performance.

Each match contains detailed statistics such as **kills, digs, assists, blocks, aces, errors, and hitting percentage**. These numbers represent the team's overall performance and help identify which skills most strongly contribute to winning.

By comparing multiple machine-learning models, the project aims to determine which model best predicts match outcomes and which factors have the greatest impact on success. The results can provide useful insights to coaches and analysts by showing which areas of play are most important for future victories.

Data & Setup

I used match-level data for the **2024** and **2025** NC State seasons. Match-level box score statistics for the 2024 and 2025 seasons were obtained from the NCAA statistics website (stats.ncaa.org) and extracted programmatically from the publicly available HTML pages. Each row is a match with team totals such as **kills, errors, total attacks, hitting percentage, assists, aces, serve errors, digs, receive attempts/errors, blocks, points,** and **ball-handling errors**. The outcome variable is **win** (1 = win, 0 = loss). During preprocessing, I removed non-match rows (such as totals, defensive totals, and header artifacts from the scraped HTML tables) so that each row in the final dataset corresponds to a single NC State volleyball match.

To keep the analysis fair for future games, I split the data by date. All matches **through Oct 11, 2025** were used for training. Matches on/after Oct 17, 2025 were reserved as a holdout test set to evaluate out-of-sample performance. For model comparison, I used repeated 5-fold cross-validation on the training set, performing the 5-fold procedure 10 times (for a total of 50 resamples). In each repetition, the data were randomly split into five parts: four folds were used for training and one for testing, rotating so each fold served as the test set once per repetition. The results were then averaged across all 50 runs to obtain more reliable and less variance-sensitive performance estimates. All numeric features were standardized (mean 0, SD 1) inside the modeling pipeline. Any missing numeric values in the training data were replaced with zeros, and the same preprocessing and standardization steps (estimated from the training set) were applied to the holdout matches to ensure all models used a consistent cleaned dataset.

Train/Test Split and Rationale

Since this project focuses on forecasting future volleyball match outcomes, the data were split chronologically rather than randomly. All available matches from the 2024 season, along with NC State's first 16 matches of the 2025 season (through October 11), were used as the **training set**. This includes 10 nonconference matches and 6 ACC matches. The remaining 14 matches from October 17 through November 29 were held out as the **test set**, representing future games the model was not allowed to see during training.

A time-based split is standard practice in sports analytics because match outcomes depend heavily on team development, lineup adjustments, roster stability, and coaching strategy—factors that evolve throughout a season. Randomly shuffling past and future games together would leak information from later matches into the training set and create an unrealistic prediction scenario.

The training period deliberately includes the entire 2024 season because it reflects the team's previous competitive level under a different roster and head coach. However, the 2025 season introduced major structural changes: the departure of many veteran players, the addition of a large freshman class, and the arrival of a new head coach implementing new strategies and rotations. Using the first 16 matches of the 2025 season in the training

set captures the team's early-season identity and ensures that model fitting reflects the group's current style of play.

The test set therefore consists of the final 14 matches of the 2025 season—matches that occur strictly after all training data. This structure avoids data leakage, respects the temporal order of the season, and evaluates how well the model can generalize to the most recent portion of the schedule, where player development, lineup adjustments, and coaching adaptations are especially important.

Model Types: Parametric vs. Non-Parametric

In this project, the four models considered differ in whether they are **parametric** or **non-parametric**.

A **parametric model** assumes a fixed functional form with a set number of parameters that do not grow with the size of the data. Logistic Regression is parametric because it assumes a linear relationship between the predictors and the log-odds of winning; once the coefficients are estimated, the model form is fully determined.

In contrast, **non-parametric models** make fewer structural assumptions and allow model complexity to grow with the data. k-Nearest Neighbors (kNN) is non-parametric because the classification boundary is determined directly by the training data, and the model can become more flexible as more observations are added. Decision Trees are also non-parametric because the splitting structure is data-driven rather than predetermined, and the tree can adaptively grow deeper depending on patterns in the predictors. Random Forests are non-parametric as well; they average many data-adaptive decision trees, allowing for highly flexible nonlinear relationships without imposing a fixed equation.

Using a mix of parametric and non-parametric models allows comparison between simpler, more interpretable structures (Logistic Regression) and more flexible, pattern-capturing approaches (Trees, Random Forests, kNN) to determine which best predicts volleyball match outcomes.

Inference vs. Prediction: Can Each Model Be Used for Inference?

Different models support inference to varying degrees. Inference refers to explaining how individual predictors relate to the outcome—such as quantifying whether higher hitting percentage increases the probability of winning. Prediction, in contrast, focuses strictly on forecasting outcomes accurately. The four models used in this project differ substantially in their ability to support formal inference.

Logistic Regression — Suitable for Inference

Logistic Regression is the only model in this analysis that supports traditional statistical inference. Its coefficients represent changes in the log-odds of winning for a one-unit increase in each predictor (holding other variables constant). This structure allows calculation of p-values, confidence intervals, and effect sizes. For volleyball data, this means logistic regression can quantify how variables such as aces, hitting percentage, kills, blocks, or digs individually influence win probability, even if its predictive performance is lower than the nonlinear models.

k-Nearest Neighbors (kNN) — Not Suitable for Inference

kNN is a purely predictive method and does not provide coefficients, p-values, or interpretable effect estimates. Predictions are determined entirely by the class labels of the closest training observations. Because there is no underlying model equation, kNN cannot tell us how strongly hitting percentage or errors impact the outcome; it can only classify based on similarity to past matches.

Decision Tree — Limited Interpretability, Not Formal Inference

Decision Trees provide a visual and intuitive flowchart structure, and individual splits can be interpreted as simple rules (e.g., “If hitting percentage > X, predict win”). However, decision trees do not support formal hypothesis testing or confidence intervals. Their interpretability is descriptive rather than inferential: they show *which thresholds* and *variables* matter for classification, but cannot quantify uncertainty or provide statistical significance.

Random Forest — Not Suitable for Inference

Random Forests are powerful predictive models but cannot be used for traditional

inference. They aggregate hundreds or thousands of trees, each using different subsets of variables and observations. Although Random Forests provide measures such as variable importance, they do not produce meaningful coefficients, p-values, or interpretable effect sizes. Their strength lies in capturing nonlinear relationships and interaction effects rather than explaining individual variable contributions in a formal inferential sense.

Summary

In this project, Logistic Regression is the only model that supports formal inference. Decision Trees and Random Forests offer interpretability through variable importance and splitting structure, but not statistical inference. kNN provides neither interpretability nor inference, serving primarily as a predictive benchmark model. This distinction highlights the trade-off between explanation (Logistic Regression) and predictive power (Random Forest, kNN, Decision Tree) within volleyball match outcome modeling.

Variable Selection

Each of the four models differs in how it handles variable selection. Variable selection refers to the model's ability to identify and retain only the most relevant predictors while excluding or down-weighting less important ones. Some models perform this process explicitly, while others rely on the full predictor set without any built-in mechanism for selecting variables.

Logistic Regression — No Built-In Variable Selection

Standard Logistic Regression does not perform variable selection automatically. All predictors included in the model contribute to the estimated coefficients unless penalization (such as LASSO or Ridge) is applied, which was not used in this project. As a result, logistic regression uses every variable in the dataset, even if some contribute little to predictive accuracy. This makes the model easier to interpret but less adaptive to redundant or weak predictors.

k-Nearest Neighbors (kNN) — No Variable Selection

kNN treats all predictors equally and computes distances in the full feature space. Because the prediction depends on similarity to neighboring observations, the method cannot ignore or down-weight irrelevant variables. In fact, adding more predictors can

degrade performance for kNN (the curse of dimensionality) by making all observations appear more distant. Thus, kNN does not perform variable selection and relies heavily on careful preprocessing and standardized features.

Decision Tree — Performs Implicit Variable Selection
Decision Trees perform *automatic* variable selection through their splitting process. At each node, the algorithm evaluates all predictors and selects the variable and threshold that best reduce impurity (e.g., Gini index). Variables that do not improve the model are never used in splits. As a result, a trained tree uses only a subset of the predictors.

Random Forest — Performs Implicit Variable Selection Through Aggregation
Random Forests also perform implicit variable selection, but through a more robust ensemble structure. Because each tree is trained on a bootstrap sample and considers only a random subset of predictors at each split (controlled by *mtry*), variables compete repeatedly across thousands of splitting opportunities. Predictors that consistently improve classification appear in more splits and receive higher importance scores, while weaker predictors contribute little. In this project, variables such as hitting percentage, aces, and kills repeatedly emerged as the strongest contributors. Although Random Forests do not perform explicit feature elimination, the ensemble structure effectively highlights the most influential predictors.

Summary
Logistic Regression and kNN do not perform variable selection and rely on the full predictor set. Decision Trees and Random Forests perform implicit variable selection by choosing splits based on predictive utility. In the volleyball dataset, the random forest naturally emphasized the most influential performance metrics—particularly hitting percentage, aces, and kills—while deemphasizing less predictive variables. This distinction is important when comparing how interpretable the models are and how flexibly they adapt to the underlying patterns in the volleyball data.

Standardization Requirements

The four models in this project differ in how they respond to predictor scaling. Standardization refers to transforming each numeric predictor to have mean 0 and

standard deviation 1. This transformation is essential for distance-based models and beneficial for any method sensitive to differences in variable scale. Because the volleyball dataset contains predictors on very different scales—for example, total attacks may exceed 150 while hitting percentage is between 0.000 and 1.000—understanding which models require standardization is important for stable and meaningful model training.

Logistic Regression — Standardization Recommended

Although Logistic Regression can be fit without standardization, it is generally recommended when predictors differ substantially in scale, as is the case with volleyball performance metrics. Standardization helps stabilize coefficient estimation, improves numerical convergence, and prevents large-scale variables from dominating the optimization process. Because this project used standardized numeric features within the modeling pipeline, the logistic regression model was fit under consistent and numerically stable conditions.

k-Nearest Neighbors (kNN) — Standardization Required

kNN relies directly on Euclidean distance computations to determine the nearest neighbors. Without standardization, variables with larger numerical ranges (e.g., total attacks, total digs) overshadow smaller-scale variables (e.g., aces, block solos). This can distort distance measures and degrade classification accuracy. Therefore, standardization is essential for kNN to ensure all predictors contribute proportionally to similarity calculations.

Decision Tree — Standardization Not Needed

Decision Trees split predictors based on threshold values that depend only on ordering, not magnitude. For example, a tree may split on “hitting percentage > 0.220” or “digs < 35,” regardless of the relative scale of the variables. Because tree-based methods do not use distance metrics or coefficient optimization, they are inherently scale-invariant. Thus, standardization does not affect their structure or performance.

Random Forest — Standardization Not Needed

Random Forests inherit the scale-invariance of individual decision trees. Since each tree constructs splits based on variable thresholds, the ensemble does not depend on the magnitude or units of the predictors. Random Forests therefore do not require any

standardization, although standardization within the preprocessing pipeline does not harm performance.

Summary

Standardization is essential for kNN and recommended for Logistic Regression, due to their sensitivity to differences in predictor scale. In contrast, Decision Trees and Random Forests do not require standardization because their splitting criteria are invariant to the units and magnitudes of the predictors. In this project, all numeric predictors were standardized through a unified preprocessing pipeline to keep the inputs consistent across models, even though Decision Trees and Random Forests do not require standardization.

Model Selection and Hyperparameter Choices

Four models were compared using repeated 5-fold cross-validation (10 repeats, 50 total resamples): Logistic Regression, k-Nearest Neighbors (k = 3), Decision Tree, and Random Forest. Performance was evaluated using Accuracy, Cohen's Kappa (κ), and ROC AUC. Logistic Regression, in its baseline glm form, does not have hyperparameters to tune. The kNN model has one hyperparameter, **k**, which was fixed at 3 for this project rather than tuned across multiple values. The Decision Tree model has hyperparameters such as maximum depth and minimum node size; these were intentionally set to **tree_depth = 4** and **min_n = 5** to provide a simple, interpretable structure suitable for a small dataset. The Random Forest model includes several hyperparameters—most notably **trees**, **mtry**, and **min_n**—and these were set to **trees = 2000**, **mtry = 4**, and **min_n = 5** based on reasonable defaults that balance flexibility, stability, and computational cost. Repeated cross-validation was then used to evaluate these fixed model specifications and compare their predictive performance, rather than performing an exhaustive tuning search. Under the repeated-CV setup, the **Decision Tree** achieved **Accuracy = 0.81**,

Kappa = 0.62, and **ROC AUC = 0.85**, while the **Random Forest** achieved **Accuracy = 0.77**, **Kappa = 0.53**, and **ROC AUC = 0.90**, slightly outperforming the other models in overall discrimination ability.

Because of its strong and consistent **AUC performance**, the **Random Forest** was selected as the **primary model** for this project. Its robustness and ability to capture nonlinear relationships among features such as **kills, digs, aces, blocks, assists, errors, and hitting percentage** make it particularly well-suited for predicting volleyball match outcomes compared to simpler linear models like Logistic Regression.

Model Evaluation (Repeated 5-Fold Cross-Validation)

We compared four models using repeated stratified 5-fold cross-validation (10 repeats, 50 total resamples) to produce stable and generalizable estimates of model performance:

Logistic Regression (engine = glm): Accuracy = 0.718, Kappa = 0.426, ROC AUC = 0.748

kNN (k = 3): Accuracy = 0.782, Kappa = 0.556, ROC AUC = 0.855

Decision Tree (depth = 4, min_n = 5): Accuracy = 0.811, Kappa = 0.615, ROC AUC = 0.823

Random Forest (trees = 2000, mtry = 4, min_n = 5): Accuracy = 0.769, Kappa = 0.529, ROC AUC = 0.903

Takeaways:

The Decision Tree achieved the highest Accuracy (0.811) and Kappa (0.615), suggesting strong agreement and effective rule-based classification, though its AUC (0.823) indicates less smooth class separation across thresholds.

The kNN (k = 3) model performed well on both Accuracy and AUC (0.855), reflecting solid ranking ability in predicting match outcomes.

The Random Forest showed the highest ROC AUC (0.903)—the best discrimination between wins and losses—while maintaining consistent overall performance, making it more robust to variation across folds. Its ensemble nature also allows identification of key volleyball metrics contributing most to match outcomes.

Logistic Regression, as expected, trailed behind the nonlinear models, capturing only the simpler linear relationships present in the data.

Additional consistency check:
Across the 10 repeated CV runs, the Random Forest's predicted win probabilities averaged about 0.32 for actual losses and 0.74 for actual wins, confirming stable class separation and consistent calibration across repetitions.

Choice for the report:
To balance classification correctness and ranking ability (AUC), the Random Forest (2000 trees, mtry = 4, min_n = 5) is selected as the primary model for interpretation and future prediction. It showed the strongest ROC AUC under repeated cross-validation and provides useful variable-importance summaries for volleyball performance metrics. On the 14-match holdout set, Logistic Regression achieved slightly higher accuracy and Kappa, but given the small sample size and the Random Forest's superior cross-validated discrimination, I treat the Random Forest as the overall "best" model, with Logistic Regression and the Decision Tree serving as simpler benchmark models for comparison.

Holdout Test Performance (2025 Late-Season Matches)

In addition to cross-validation on the training set, all four models were evaluated on the chronological holdout set: the final 14 matches of the 2025 season (October 17–November 29). This provides an out-of-sample check on how well each model generalizes to late-season games that were not used during training.

Holdout performance (accuracy, Cohen's Kappa, and ROC AUC) was:

- **Logistic Regression (engine = glm)**: Accuracy = 0.857, Kappa = 0.696, ROC AUC = 0.188
- **kNN (k = 3)**: Accuracy = 0.786, Kappa = 0.571, ROC AUC = 0.146

- **Decision Tree (depth = 4, min_n = 5)**: Accuracy = 0.714, Kappa = 0.417, ROC AUC = 0.375
- **Random Forest (trees = 2000, mtry = 4, min_n = 5)**: Accuracy = 0.786, Kappa = 0.571, ROC AUC = 0.125

On this 14-match holdout set, Logistic Regression achieved the highest accuracy and Kappa. The non-parametric models—kNN, the Decision Tree, and the Random Forest—performed slightly worse but still generalized reasonably well given the limited size of the test set.

Taken together, the cross-validation and holdout results suggest that:

- Decision Trees and Random Forests provide strong discrimination on the training-period matches (higher CV AUC),
- Logistic Regression performs surprisingly well on the final 14 matches, despite being a simpler parametric model, and
- all four models exhibit broadly similar accuracy on the holdout set, with differences that could easily be due to sampling variability in such a small test window.

Variable Importance & Interpretation

To understand which performance metrics most strongly influenced match outcomes, we examined **variable importance** from the **Random Forest model**. Importance values reflect how much each variable reduced classification error when included in tree splits across the 2000 trees.

The top predictors were:

- **Hitting Percentage (importance = 4.25)** – by far the most influential variable, aligning with its role as a key indicator of offensive efficiency. Teams with higher hitting percentages generally demonstrate stronger attack consistency and shot selection.

- **Aces (3.37)** – serving strength strongly impacts match momentum and point scoring; high ace counts often reflect aggressive, well-targeted serving.
- **Kills (1.90)** and **Points (1.86)** – both are direct measures of offensive success, capturing how well a team converts attacks into points.
- **Errors (1.52)** and **Receive Attempts (1.35)** – these provide balance between offense and defense, indicating how often a team keeps rallies alive or loses possession.
- **Assists (1.25)** and **Digs (1.17)** – setters and defensive specialists play an essential role in sustaining rallies and enabling transitions between offense and defense.

While **blocking metrics** (Block Solos = 0.75; Block Assists = 0.45) ranked lower than expected, this may reflect that blocking outcomes are already captured indirectly by **opponent errors** and **hitting percentage**. Since a successful block lowers the opponent's hitting efficiency, its effect can appear in other correlated variables rather than in raw block counts alone.

Overall, the model suggests that **offensive consistency (hitting %, kills, aces)** and **serve-receive control (errors, receive attempts)** drive match outcomes more strongly than raw block counts. This insight aligns with collegiate-level volleyball trends, where precision and efficiency across multiple touches often outweigh single defensive events.

The Random Forest variable-importance chart (rf_variable_importance.png) visually reinforces these findings, with a steep decline in contribution after the top five metrics, confirming that performance prediction is dominated by attacking and serving efficiency.

After completing model evaluation, I refit the Random Forest on the full dataset (training plus holdout matches) and recomputed variable importance; the ranking of predictors was nearly identical, confirming that hitting percentage, aces, points, kills, and receive attempts remained the dominant drivers of match outcomes.

Conclusion & Next Steps

This project successfully built and evaluated multiple machine learning models to predict NC State Women's Volleyball match outcomes using comprehensive team performance data. Repeated 5-fold cross-validation showed that the **Random Forest model** achieved the strongest overall discrimination ability (ROC AUC = 0.903) and competitive accuracy, making it the best performer during training. The **holdout evaluation**, however, demonstrated that **Logistic Regression generalized similarly well** on the final 14 matches of the 2025 season, achieving the highest accuracy among the four models. This indicates that even a simpler linear model can perform well on unseen matches, especially when the underlying relationships remain stable across the season.

The Random Forest's **ensemble structure** allowed it to capture complex nonlinear relationships involving hitting percentage, kills, aces, blocks, and errors—metrics that together reflect offensive efficiency and defensive stability. Consistent with volleyball analytics research, the model emphasized that attacking efficiency and serve-receive control are more predictive of match outcomes than isolated defensive statistics such as raw block counts. Blocking contributed to predictions, but its influence was smaller relative to variables that more directly capture offensive momentum and rally outcomes.

Overall, the set of models demonstrated **reasonable generalization** to new data. Accuracy values on the holdout matches remained close to their cross-validation estimates, and no model showed signs of extreme overfitting. Because the test set contained only 14 matches, some metrics such as ROC AUC showed higher variability, but the accuracy and kappa values indicate stable performance across models.

Moving forward, future extensions could incorporate additional contextual and opponent-adjusted variables—such as opponent strength, match location, lineup availability, or player-level performance—to further refine predictions. Incorporating richer information may help identify situational factors that influence match outcomes and could provide a more complete, data-driven framework for supporting coaches, analysts, and fans in understanding NC State's competitive performance.